

QUESTION TWO

- a. If $p = 3 - i$ and $q = -2 + 5i$, find $\bar{p} - 3q$, giving your solution in the form $a + bi$.

$$\begin{aligned}\bar{p} - 3q &= 3 + i - 3(-2 + 5i) \\ &= 3 + i + 6 - 15i \\ &= 9 - 14i\end{aligned}$$

- b. Write $\frac{3}{4-\sqrt{5}}$ in the form $a + b\sqrt{5}$ where a and b are rational numbers.

$$\begin{aligned}\frac{3}{4-\sqrt{5}} &\times \frac{4+\sqrt{5}}{4+\sqrt{5}} \\ &= \frac{12+3\sqrt{5}}{11}\end{aligned}$$